

From Trade War to Green Transition

Optimal Electric Vehicle Tariffs with Revenue-Funded Subsidies

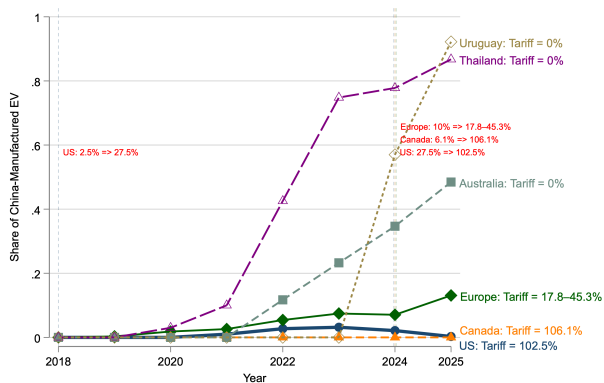
Panle Jia Barwick Jack Collison Pinelopi K. Goldberg
Shanjun Li Yucheng Wang

Presenter: Ruoyu Men

September 18, 2026

NBER Working Paper 35334 · June 2026 version

Can trade policy support the green transition?



Can access to affordable EVs coexist with support for domestic producers?

Fig. 1(b), p.35.

A joint policy problem

- Access to Chinese EVs, domestic industry, the environment, and public finance.
- Choose tariffs and domestic EV subsidies together.

Theory disciplined by a global automobile model

- Differentiated products, heterogeneous consumers, multiproduct pricing.
- Estimates from 13 countries; counterfactual prices, quantities, and welfare.

Main answer: moderate tariffs with revenue-funded subsidies can approach the unconstrained optimum in the estimated markets.

Sec. 1, pp.3–6.

Related literature

Literature	This paper's connection
Green industrial policy Head et al. (2026)	Jointly evaluate trade, domestic support, environmental benefits and fiscal costs.
Auto trade under oligopoly Goldberg (1995); Berry–Levinsohn–Pakes (1999)	From auto export restraints and exchange-rate pass-through to tariffs with revenue-funded subsidies.
EV adoption and subsidies Springel (2021)	Design a domestic subsidy together with the tariff that finances it.

Sec. 1, pp.7–8.

1. Theoretical model
2. Background and data
3. Empirical model, identification and estimates
4. Counterfactual framework
5. Welfare effects of specified US and EU policies
6. Optimal policy design
7. Conclusion and discussion

Secs. 2–8.

Demand and firm pricing

$$q_j = q_j(\mathbf{p}), \quad \max_{\{x_j: j \in \mathcal{J}_f\}} \sum_{j \in \mathcal{J}_f} (x_j - c_j) q_j(\mathbf{p}).$$

$\mathcal{J}_f$: firm f 's portfolio; q_j : sales; c_j : marginal cost.

Tariff on imports $j \in \mathcal{C}$

$$p_j = (1 + \tau)x_j$$

τ : ad valorem tariff rate.

x_j : **firm receipts.** p_j : **buyer payment.**

Other products: $p_j = x_j$. Firms choose prices given rivals' prices.

Sec. 2, p.8.

Eligible EV subsidy $j \in \mathcal{D}$

$$p_j = x_j - b$$

b : subsidy per vehicle.

Pricing and pass-through

$$x_j = c_j + \underbrace{m_j(\mathbf{p}; \tau, b)}_{\text{endogenous markup: demand, ownership and policy wedges}}$$

$$\frac{dp_j}{d\tau} = \underbrace{x_j}_{\text{direct tax wedge}} + \underbrace{(1 + \tau) \frac{dx_j}{d\tau}}_{\text{firm repricing}}, \quad j \in \mathcal{C}$$

$$\frac{dp_j}{db} = \underbrace{\frac{dx_j}{db}}_{\text{firm repricing}} - \underbrace{1}_{\text{direct subsidy}}, \quad j \in \mathcal{D}$$

Demand and competition determine how much of the policy reaches buyers.

Vary one instrument, holding the other fixed: a tax *rate* or a unit subsidy.

Sec. 2, pp.8–9; Eq. (B.3), p.53.

Welfare and policy choice

$$\max_{\tau, b \geq 0} W(\tau, b), \quad W = \underbrace{CS}_{\text{consumers}} + \underbrace{\Pi_{\text{dom}}}_{\text{domestic firms}} + \underbrace{EB}_{\text{environment}} + \underbrace{G}_{\text{public finance}}.$$

	Tariff	Subsidy
Consumers	Import prices and substitution	Effective prices and product choice
Domestic profits	Diversion from foreign firms	Sales, markups and ICE cannibalization
Environment	EV/ICE substitution	Avoided ICE purchases
Public finance	Revenue and the tax base	Expenditure and both fiscal bases

Sec. 2.1, Eq. (3), pp.9–10.

Joint optimum

$$\max_{\tau, b \geq 0} W(\tau, b)$$

Allows a fiscal surplus or deficit.

Revenue recycling

$$\max_{\tau, b \geq 0} W(\tau, b) \quad \text{s.t. } G = 0$$

Tariff receipts finance EV subsidies.

$$G = \tau \sum_{j \in \mathcal{C}} x_j q_j - b \sum_{j \in \mathcal{D}} q_j.$$

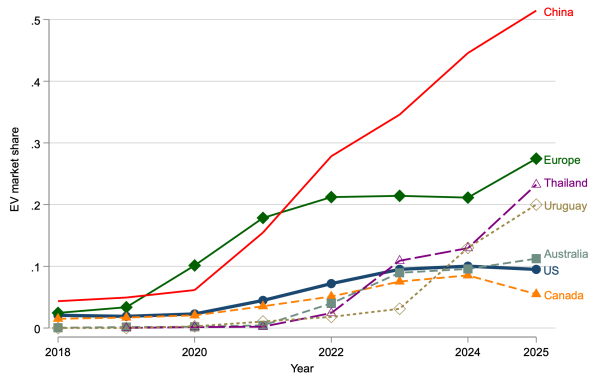
$$V = W - G, \quad \mathcal{L} = W + \mu G = V + \lambda G, \quad \lambda = 1 + \mu.$$

V : nonfiscal welfare; μ : added constraint multiplier; λ : total fiscal weight.

Near a regular interior optimum, small budget-driven policy adjustments have a second-order welfare cost.

Sec. 2.2, p.11.

Background: Chinese EVs



EV adoption accelerates while market access and vehicle prices differ across countries.

Sec. 3.1, p.12; Fig. 1(a), p.35.

13 countries · 2004–2023 · 65,574 model–country–year observations

Objects	Sources / content	Role in the model
Sales and prices	MarkLines; CATARC/IHS; price sources	Shares and pricing
Vehicle attributes	Powertrain, range, power, footprint; battery suppliers	Substitution and cost exposure
Income	Population income; buyer surveys	Price sensitivity and buyer sorting
Policy and markets	Taxes, tariffs, EV subsidies; competing model menus	Price wedges and identification

EV = BEV + PHEV; ICE = internal-combustion-engine vehicles.

Sec. 3.2, pp.12–13; Table 1, p.41; fn. 2, p.4.

$$u_{ijmt} = -\alpha_{im}p_{jmt} + X'_{jmt}\beta_{im} + \eta_{jmt} \\ + \sigma_m^{\text{brand}}\nu_{i\ell(j)m} + \xi_{jmt} + \varepsilon_{ijmt}.$$

i, j, m, t

Consumer, model, country market, year

$p_{jmt}; \alpha_{im}$

Net buyer payment; price sensitivity

$X_{jmt}; \beta_{im}$

Observed attributes; heterogeneous tastes

$\eta_{jmt}; \xi_{jmt}$

Fixed effects; unobserved quality

$\ell(j); \nu_{i\ell(j)m}$

Brand; shared consumer–brand draw

ε_{ijmt}

i.i.d. Type I extreme value shock

Outside option: $u_{i0mt} = \varepsilon_{i0mt}$, no purchase of an inside new car.

Sec. 4.1, Eq. (4), p.14.

Heterogeneity and substitution

Income and price sensitivity

$$\alpha_{im} = \bar{\alpha} + \frac{\alpha_{g(m)}}{y_{im}} + \sigma_p \nu_{im}^p.$$

y : income; $g(m)$: four country groups; ν^p is log-normal.

One brand draw shared across the consumer's choices

$$\sigma_m^{\text{brand}} \nu_{il(j)m}, \quad \beta_{imk} = \bar{\beta}_k + \sigma_m^k \nu_{im}^k.$$

Brand and attribute draws are normal. Brand loyalty links EV and ICE substitution.

$$s_{ijmt} = \frac{e^{v_{ijmt}}}{1 + \sum_{k \in \mathcal{J}_{mt}} e^{v_{ikmt}}}, \quad q_{jmt} = M_{mt} \int s_{ijmt} dF_m(i).$$

$v = u - \varepsilon$; M : potential market size; F : consumer income/taste distribution.

Sec. 4.1, Eqs. (4)–(5), pp.14–15.

$$\max_{\{x_j: j \in \mathcal{J}_f\}} \Pi_f = \sum_{k \in \mathcal{J}_f} (x_k - c_k) q_k(\mathbf{p}).$$

$$\underbrace{q_j}_{\text{revenue on existing sales}} + a_j \underbrace{\sum_{k \in \mathcal{J}_f} (x_k - c_k) \frac{\partial q_k}{\partial p_j}}_{\text{quantity effects across the owned portfolio}} = 0.$$

$a_j = \partial p_j / \partial x_j$: $1 + \tau$ for taxed imports, 1 otherwise.

Demand estimates $\longrightarrow$ markups $\longrightarrow$ marginal costs

$$\hat{\mathbf{c}} = \mathbf{x}^{obs} - \hat{\mathbf{m}}^{obs}.$$

Diversion to another owned model offsets part of the own-model sales loss.

Sec. 4.2, Eqs. (6)–(7), pp.15–16.

$$\mathbb{E}[\xi_{jmt} \mid Z_{jmt}, X_{jmt}] = 0.$$

Instruments	Source of price variation
BLP product-menu IVs	Own/rival brand counts, model counts and attributes shift markups.
Local differentiation	Counts within one standard deviation measure nearby substitutes.
Battery capacity $\times$ supplier	Upstream supply exposure shifts costs.

Local scope: country $\times$ year $\times$ segment $\times$ fuel type.

Exclusion: conditional on controls, instruments are orthogonal to remaining quality.

Sec. 4.3, p.16.

Population income is an input; buyer income is a model outcome.

$$\mathbb{E}[y_i \mid j] = \frac{\int y_i s_{ij} dF(i)}{\int s_{ij} dF(i)} \longleftrightarrow \text{survey mean for model } j.$$

124 moments	Mean buyer income: 45 models sold in China (2018–20); 79 sold in the US (2018).
21 moments	EV-buyer income shares in six other countries: 27 bins less six adding-up restrictions.

Two-step GMM: fit shares and moments, update moment weights, re-estimate.

Sec. 4.3, p.16.

Demand estimates

Preferred specification: country $\times$ year $\times$ EV and brand $\times$ EV fixed effects

Parameter	Estimate	SE
Home-brand indicator	0.111	0.034
EV taste dispersion	2.485	0.109
Brand dispersion: US and Canada	1.568	0.126
Brand dispersion: China	0.252	0.438
Income-price term: CA/NO/US/CH	1.107	0.074
Price dispersion	0.103	0.006

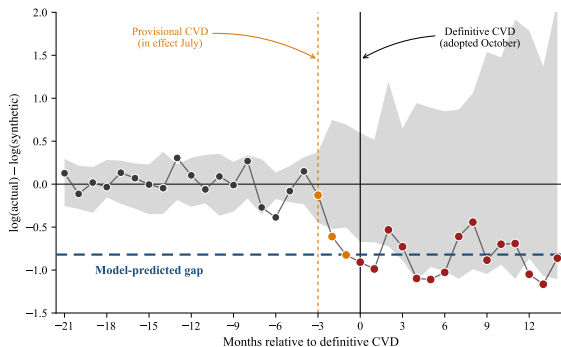
Heterogeneity disciplines substitution and the distribution of policy gains.

Average own-price elasticity: -2.62 . Standard errors clustered by country-brand.

Sec. 4.4, p.17; Table 1, col. (4), p.41.

Out-of-sample sales response

2024 EU countervailing duties (CVD): a post-estimation sales check



Model: -56.0% · Table 3's 2023 Chinese-owned menu.

Synthetic control: -57.4% · Five-brand incumbent BEV/PHEV sample in EU-6.

Sec. 4.4, pp.17–18; App. D.1, p.67; Fig. E.5, p.74.

Consumer surplus: individual logsum changes, in dollars

$$\Delta CS = M \int \frac{1}{\alpha_i} \log \frac{1 + \sum_{j \in \mathcal{J}^{CF}} e^{v_{ij}^{CF}}}{1 + \sum_{j \in \mathcal{J}^{SQ}} e^{v_{ij}^{SQ}}} dF(i).$$

Domestic profits and public finance

$$\Pi_{dom} = \sum_{j \in \mathcal{J}_{dom}} (x_j - c_j) q_j, \quad G = \tau \sum_{j \in \mathcal{C}} x_j q_j - b \sum_{j \in \mathcal{D}} q_j.$$

US per-ICE lifetime damage calibration

$$\Delta EB = -\Delta Q_{ICE} \times (\$3,810 + \$687).$$

CF : counterfactual; SQ : status quo. Both inside menus exclude option 0 (the leading 1). $\mathcal{J}_{dom}$: products whose profits count nationally.

Sec. 5.1, Eqs. (8)–(10), pp.18–19; App. C.3, p.63.

Introducing Chinese EVs

Ten models: 34.3% of China's 2023 EV sales

Demand appeal

- Observed Chinese vehicle attributes and unobserved quality.
- US–China fixed-effect gaps from overlapping non-Chinese models, by powertrain.
- Apply the estimated US preference distribution.

Export-market cost

- China–Europe cost analogues.
- Shipping: \$2,750 per vehicle.
- US distribution and marketing: \$1,402 per vehicle.

US demand appeal and costs are **constructed from cross-market information**.

Sec. 5.2, p.19; App. C.2, pp.61–63.

Counterfactual implementation

Fixed inputs Preferences, marginal costs and ownership.

Changed inputs Policies and the specified product menu.

$$(\tau, b, \mathcal{J}) \longrightarrow \text{all firms reprice} \longrightarrow \mathbf{q} \longrightarrow (CS, \Pi_{dom}, EB, G).$$

	United States	European Union
Benchmark	No entry / no policy	Chinese EVs present; 10% MFN tariff
Product change	Add ten constructed models	Existing menu; remove Chinese EVs in the ban experiment

Sec. 5.2, p.19; Tables 2–3, pp.42–43.

US tariff and subsidy policies

US, 2023 · Changes versus no entry · Monetary changes in \$bn

	Ban	2.5%	27.5%	102.5%	102.5% + recycle
EV share (%)	9.00	18.92	16.06	11.57	15.83
$\Delta\Pi_{US}$	0	-10.44	-8.70	-4.49	2.92
ΔG	0	1.61	11.69	14.48	0
ΔW	0	37.48	40.88	30.78	34.59

Recycling at the specified 102.5% tariff implies a \$8,139 subsidy.

Ban + \$7,500 domestic subsidy: EV share 13.08%; $\Delta W = \$5.64\text{bn}$.

Welfare rises from 2.5% to 27.5%, then falls at 102.5%; a high tariff still allows entry.

Sec. 6.1, pp.20–22; Table 2, cols. (1)–(6), p.42.

EU trade remedies

EU-6, 2023 menu · Changes versus Chinese EVs present with 10% MFN

	CVD	CVD + recycle	MPC	Remove
EV share (%)	23.11	23.74	23.11	22.54
ΔCS (\$bn)	-2.22	-0.68	-2.22	-5.64
ΔG (\$bn)	1.41	0	0	-1.15
ΔW (\$bn)	-0.11	1.23	-1.52	-5.00

Minimum price commitment: same prices and quantities as CVD, different ownership of tariff rents.

Table 3 reports zero net fiscal change under recycling: $\Delta G = 0$.

Sec. 6.2, pp.22–23; Table 3, p.43.

Optimal policy regimes

US, 2023 · Ten entrants · Changes versus no entry

	Tariff only	Subsidy only	Joint	Recycling
Tariff (%)	27.5	0	22.5	25.0
Subsidy (\$1,000/eligible EV)	0	9.60	8.40	7.47
ΔW (\$bn)	40.88	42.94	45.45	45.33
ΔG (\$bn)	11.69	-14.77	-3.01	0
EV share (%)	16.06	23.57	20.41	19.60

No-entry baseline EV share: 9.00%. EV entries are levels, not changes.

Recycling welfare: +\$45.33bn vs no entry; +\$4.45bn vs optimal tariff only;
-\$0.12bn vs unconstrained joint optimum.

Table 4, p.44.

Distributional effects

US revenue recycling · New-car EV shares by income quartile

	Baseline EV share	Recycling EV share	ΔCS (\$bn)
Q1: lowest	6.80%	19.05%	8.15
Q2	7.56%	16.04%	6.57
Q3	10.51%	20.59%	15.12
Q4: highest	11.15%	22.71%	13.26

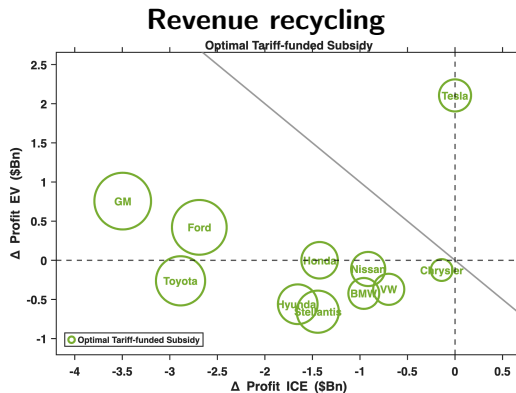
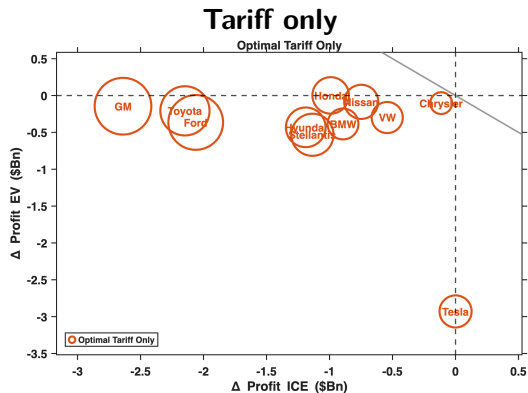
Q3 has the largest group-total consumer-surplus gain: **\$15.12bn.**

Consumer-surplus entries are group totals; EV shares are within-group new-car sales.

Sec. 7.1, p.24; Table 5, p.45.

Firm profits

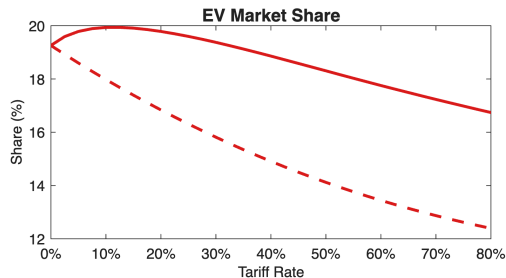
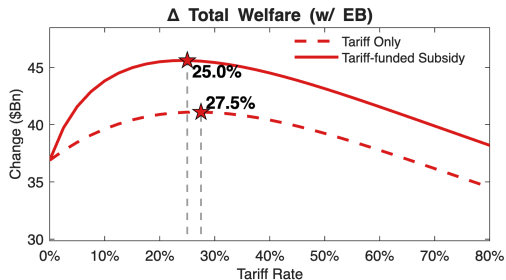
Incumbents in the US market; EV and ICE profit changes



Recycling reduces the aggregate US-firm loss from \$8.70bn to \$3.24bn, but does not eliminate it.

Sec. 7.1, p.25; Fig. 2(a,b), p.36.

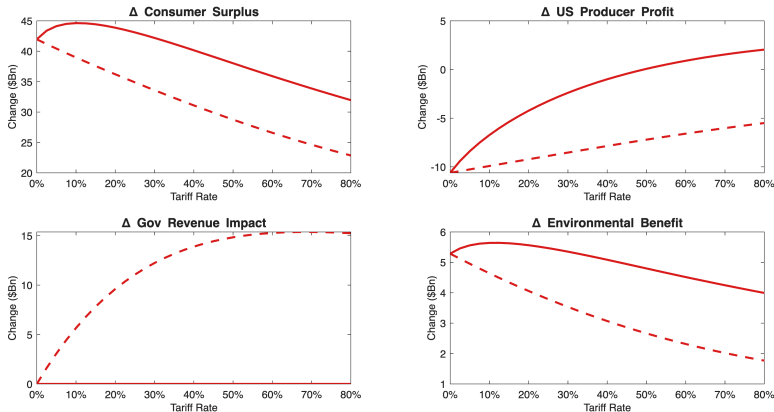
Welfare implications



Recycling raises welfare and EV adoption at most tariffs; separate optima: 25% vs 27.5%.

Sec. 7.2, pp.25–26; Fig. 3, p.37.

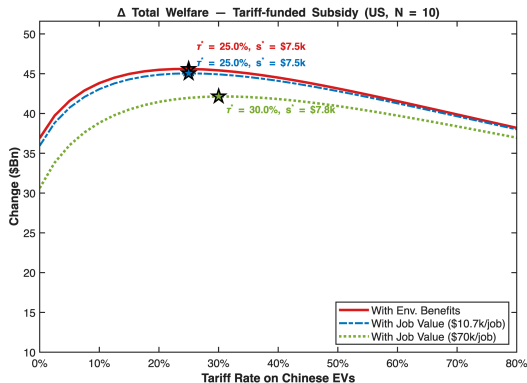
Welfare implications



Solid: recycling; dashed: tariff only. Gains to consumers, firms and the environment offset forgone fiscal surplus.

Sec. 7.2, pp.25–26; Fig. 3, p.37.

Employment and optimal policy



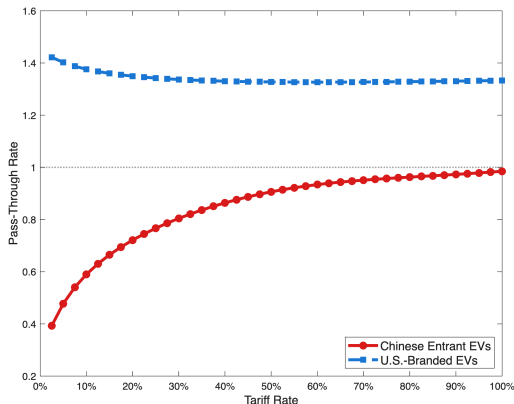
At \$70,000/job-year: optimal tariff 25% $\rightarrow$ 30%; subsidy \$7.5k $\rightarrow$ \$7.8k.

Greater job value strengthens protection; labor intensities and local-production shares stay fixed.

Sec. 7.2, pp.26–27; App. C.4, pp.63–65; Fig. E.7, p.76.

Pass-through

Model-implied group responses along the joint tariff–recycling path



Sec. 7.3, pp.27–28; Fig. 5(a), p.39.

Chinese entrant EVs

$$\Delta p / \Delta T < 1$$

Exporters absorb part of the tax.

US-branded EVs

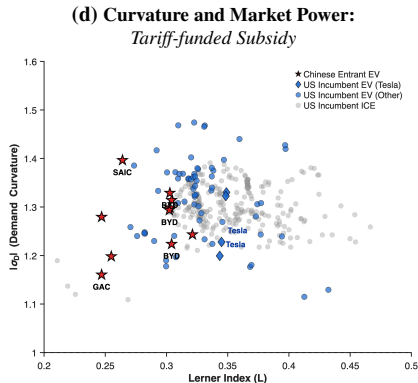
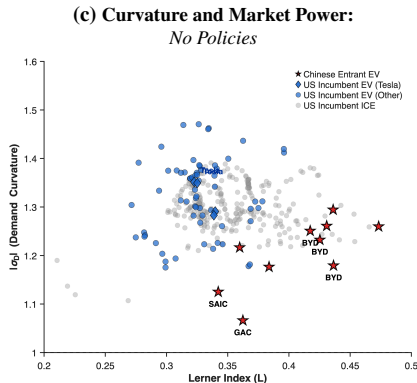
$$-\Delta p / \Delta b > 1$$

Price cuts exceed the unit subsidy.

$T_j = \tau x_j$: actual tax per car.

Markups and demand curvature

Ten entrants present in both panels



Entrants compress markups; buyer heterogeneity amplifies US EV subsidy pass-through.
Horizontal: $L = (P - MC)/P$. Vertical: absolute demand curvature, $|\sigma_d|$.

Sec. 7.3, pp.27–28; Fig. E.8(c,d), p.77.

Cross-country comparisons

Unconstrained joint optima · Common ten-entrant experiment

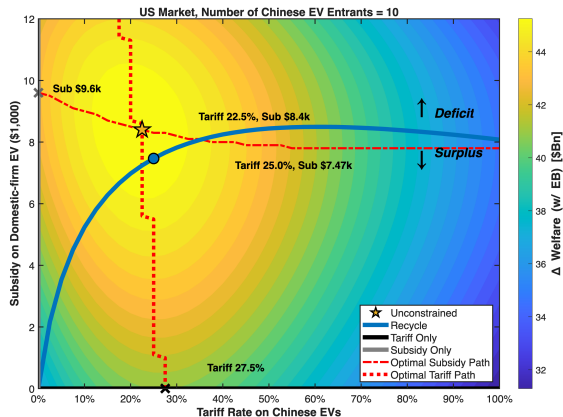
	Spain	UK	US	Germany
Home-HQ EV models	0	8	16	61
Their markup (\$1,000)	–	30.11	16.65	23.67
Optimal tariff (%)	17.5	20.0	22.5	37.5
Optimal subsidy (\$1,000)	0	12.80	8.40	10.60

Two policy motives: domestic diversion and foreign-rent extraction; correcting domestic market power.

The UK has few home models but the largest optimal unit subsidy.

Sec. 7.4, pp.28–29; Table 6, p.46.

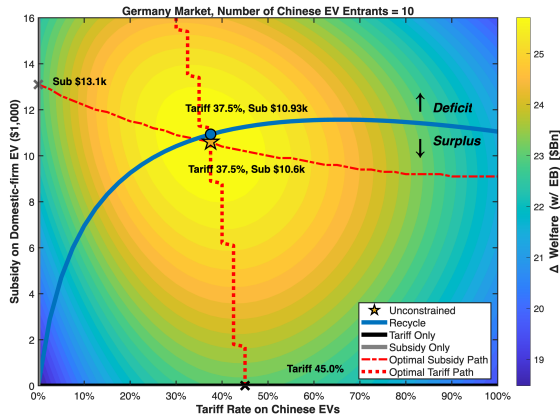
Policy mix and budget balance



At a regular interior joint optimum, $\nabla W(\gamma^*) = 0$, $\gamma = (\tau, b)$.

Sec. 7.4, pp.28–29; Fig. 6(a,b), p.40.

Policy mix and budget balance

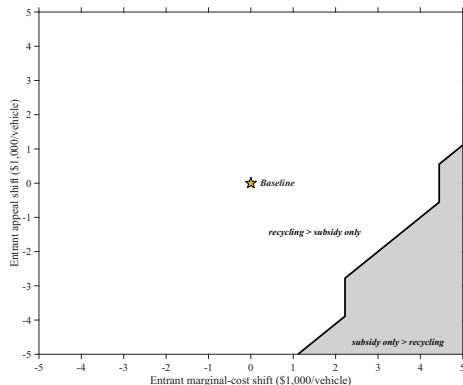


At a regular interior joint optimum, $\nabla W(\gamma^*) = 0$, $\gamma = (\tau, b)$.

Sec. 7.4, pp.28–29; Fig. 6(a,b), p.40.

Sensitivity to entrant assumptions

(a) Policy-regime regions



Re-optimize in each scenario. Subsidy-only can win when entrants are costly and less attractive.

App. D.2, p.68; Fig. E.10(a), p.79.

Can trade policy support the green transition?

- Moderate tariffs retain gains from entry; revenue-funded subsidies partly offset domestic losses.
- US example: 25% tariff + \$7,470 subsidy; EV share 9.00% → 19.60%.
- 95.44 Mt avoided **lifetime** CO₂ for the year's new-car cohort.

Market structure determines the mix.

Substitution, markups and domestic portfolios govern the gains from each instrument.

Short- to medium-run scope: product portfolios and production locations are given.

Sec. 8, pp.30–31; Table 4, p.44; App. C.3, p.63.

Production location: what happens to the tariff base?

If firms localize production, import revenue and subsidy eligibility can change together.
Head, Mayer, Melitz and Yang (2026): multi-stage production and EV industrial policy.

Revenue use: vehicles, charging, or both?

Charging investment can affect demand for both imported and domestic EVs.
Springel (2021): incentives on both sides of the EV–charging market.

How do consumers' trade-offs between upfront purchase savings and future charging benefits shape the self-financing policy mix?

Sec. 8, pp.30–31; Head et al. (2026); Springel (2021).

Keep three price objects separate

Talk symbol	Economic object	Paper's theoretical notation
x_j	Net receipts per unit sold	P_j
p_j	Consumer price after policy	$\tilde{P}_j$
$T_j = \tau x_j$	Actual tariff payment per import	τP_j

$$\mathbf{p} = A(\tau)\mathbf{x} + \mathbf{o} - b\mathbf{d}, \quad A_{jj} = 1 + \tau \mathbf{1}\{j \in \mathcal{C}\}, \quad d_j = \mathbf{1}\{j \in \mathcal{D}\}.$$

B.4: $c_j = mc_j + \text{shipment}$; $o_j = \text{other Costs}$ enters after the tariff.

For an unsubsidized import, $T_j = \tau x_j = p_j - x_j - o_j$. Define policy eligibility separately from ownership.

Sec. 2, p.8; Eqs. (B.1), (B.4)–(B.5), p.53.

A tariff can transfer foreign rents

Illustration: one imported good, no domestic producer or externality

$$W = CS(p) + \tau x q(p), \quad p = (1 + \tau)x.$$

$$\frac{dW}{d\tau} = -q \frac{dp}{d\tau} + \left(x + \tau \frac{dx}{d\tau} \right) q + \tau x q' \frac{dp}{d\tau} = -q \frac{dx}{d\tau} + \tau x q' \frac{dp}{d\tau}.$$

- If x falls, the first term is a rent-extraction gain.
- If p rises and $q' < 0$, the second term is a distortion cost.

Sec. 2, pp.8–10.

Multiproduct pricing with policy wedges

Firm-receipt derivatives and ownership-adjusted demand

$$\pi_f = \sum_{k \in J_f} (x_k - c_k) q_k(\mathbf{p}), \quad \mathbf{p} = A\mathbf{x} - b\mathbf{d}.$$

$$\frac{\partial \pi_f}{\partial x_j} = \underbrace{q_j}_{\text{direct receipts}} + \underbrace{A_{jj} \sum_{k \in J_f} (x_k - c_k) \frac{\partial q_k}{\partial p_j}}_{\text{own-portfolio quantity responses}} = 0.$$

$$\Delta_{jk}^x = -A_{jj} \frac{\partial q_k}{\partial p_j} \Omega_{jk} \Rightarrow \mathbf{q} - \Delta^x (\mathbf{x} - \mathbf{c}) = 0 \Rightarrow \mathbf{c} = \mathbf{x} - (\Delta^x)^{-1} \mathbf{q}.$$

$\Omega_{jk} = 1$ for common ownership. For two owned products, the first FOC includes both $(x_1 - c_1)q_{1,1}$ and $(x_2 - c_2)q_{2,1}$; the latter drops out under separate ownership.

Sec. 4.2, Eqs. (6)–(7), pp.15–16; App. B.1, p.53.

From pricing conditions to policy responses

Differentiate the equilibrium system, then translate receipts into buyer prices

$$F(\mathbf{x}, \gamma) = 0, \quad \gamma = (\tau, b)^\top \quad \Rightarrow \quad F_x \mathbf{x}_\gamma + F_\gamma = 0 \quad \Rightarrow \quad \mathbf{x}_\gamma = -F_x^{-1} F_\gamma.$$

$$\mathbf{p}_\tau = \underbrace{A\mathbf{x}_\tau}_{\text{equilibrium repricing}} + \underbrace{\text{diag}(\mathbf{1}_C)\mathbf{x}}_{\text{direct tax wedge}},$$

$$\mathbf{p}_b = \underbrace{A\mathbf{x}_b}_{\text{equilibrium repricing}} - \underbrace{\mathbf{d}}_{\text{direct subsidy wedge}}.$$

F_x : slopes, curvature, ownership and price interactions. Requires a differentiable local branch and nonsingular F_x .

Eq. (2), p.9; App. B, pp.53–59.

Deriving subsidy overshifting step by step

Single product; fixed rivals; constant cost; $q > 0$, $q' < 0$; regular interior optimum

1. FOC: $q + (p + b - c)q' = 0 \Rightarrow p = c - b + m(p), \quad m = -q/q'.$

2. Markup slope: $m'(p) = -\frac{(q')^2 - qq''}{(q')^2} = \rho - 1, \quad \rho = \frac{qq''}{(q')^2}.$

3. Total derivative: $p_b = -1 + (\rho - 1)p_b \Rightarrow -p_b = \frac{1}{2 - \rho}.$

4. SOC: $\pi_{pp} = q'(2 - \rho) < 0 \Rightarrow \rho < 2; \quad -p_b > 1 \iff 1 < \rho < 2.$

For example, $\rho = 1.5$ gives local pass-through of 2. Convexity alone is insufficient; the normalized curvature and second-order condition both matter.

App. B.2, pp.54–59.

Ad valorem tax incidence is different

Single product; $p = (1 + \tau)x$ and $T = \tau x = p - x$

$$\max_p \left(\frac{p}{1 + \tau} - c \right) q(p) \implies p - (1 + \tau)c = -\frac{q}{q'}.$$

$$\frac{dp}{d\tau} = \frac{c}{2 - \rho}, \quad \frac{dT}{d\tau} = \frac{p}{(1 + \tau)^2} + \frac{\tau}{1 + \tau} \frac{dp}{d\tau}.$$

$$\frac{dp}{dT} = \frac{c(1 + \tau)^2}{p(2 - \rho) + \tau c(1 + \tau)}.$$

With a positive denominator, $dp/dT > 1$ iff $\rho > 2 - (1 + \tau)c/p$. This is not the subsidy threshold $\rho > 1$.

App. B.2, pp.55–59.

The budget curve has endogenous fiscal bases

Follow the policy path that keeps tariff receipts equal to subsidy expenditure

$$R(\tau, b) = \sum_{j \in \mathcal{C}} x_j q_j, \quad S(\tau, b) = \sum_{j \in \mathcal{D}} q_j, \quad G(\tau, b) = \tau R - bS.$$

$$\underbrace{G_\tau = R + \tau R_\tau - bS_\tau}_{\text{tax rise and both fiscal-base responses}}, \quad \underbrace{G_b = \tau R_b - S - bS_b}_{\text{subsidy rise and both fiscal-base responses}}.$$

$$G(\tau, b(\tau)) = 0 \quad \Rightarrow \quad G_\tau + G_b b'(\tau) = 0 \quad \Rightarrow \quad b'(\tau) = -\frac{G_\tau}{G_b}.$$

$$\left. \frac{dW}{d\tau} \right|_{G=0} = W_\tau + W_b b'(\tau) = W_\tau - W_b \frac{G_\tau}{G_b}.$$

Requires $G_b \neq 0$. If $G_\tau > 0$ and $G_b < 0$, a higher tariff finances a larger unit subsidy. The path need not remain upward-sloping everywhere.

Sec. 2.2, p.11; Eq. (9), p.18; App. A, pp.49–52.

Why the local welfare cost is second order

At a regular interior optimum γ^* : $H = -\nabla^2 W \succ 0$ (2×2); $g = \nabla G \neq 0$ (2×1)

1. Welfare: $\nabla W(\gamma^*) = 0 \Rightarrow W(\gamma^* + h) = W^* - \frac{1}{2}h^\top H h + o(\|h\|^2).$

2. Budget: $g^\top h \simeq \Delta G, \quad \Delta G = \bar{G} - G(\gamma^*).$

3. Adjustment: $\min_h \frac{1}{2}h^\top H h \text{ s.t. } g^\top h = \Delta G \Rightarrow Hh = \eta g, \quad h = \eta H^{-1}g.$

4. Loss: $\kappa = \frac{1}{g^\top H^{-1}g}, \quad \eta = \kappa \Delta G, \quad W^* - W^c \simeq \frac{1}{2}\kappa(\Delta G)^2.$

$h \in \mathbb{R}^2$: displacement; η : quadratic-program multiplier; κ : the paper's curvature in the budget direction. Curvature governs cost; corners and distant constraints need separate analysis.

App. A.1, Eqs. (A.1)–(A.2), pp.51–52.

Heterogeneity enters demand derivatives

$$s_{ij} = \frac{\exp(v_{ij})}{1 + \sum_k \exp(v_{ik})}, \quad q_j = M \int s_{ij} dF(i).$$

$$\frac{\partial q_j}{\partial p_j} = -M \int \alpha_i s_{ij} (1 - s_{ij}) dF(i),$$

$$\frac{\partial q_j}{\partial p_k} = M \int \alpha_i s_{ij} s_{ik} dF(i) \quad (k \neq j).$$

$$\frac{\partial^2 q_j}{\partial p_j^2} = M \int \alpha_i^2 s_{ij} (1 - s_{ij}) (1 - 2s_{ij}) dF(i).$$

The relevant population is the whole choice distribution, including the outside option. Different consumer types enter and leave as prices move.

Sec. 4.1, Eq. (5), p.15.

What exactly do the micro moments match?

$$\mathbb{E}_{\theta}[y_i \mid j] = \frac{\int y_i s_{ij}(\theta) dF(i)}{\int s_{ij}(\theta) dF(i)}.$$

$$\Pr_{\theta}(y_i \in B \mid EV) = \frac{\int \mathbf{1}\{y_i \in B\} \sum_{j \in EV} s_{ij}(\theta) dF(i)}{\int \sum_{j \in EV} s_{ij}(\theta) dF(i)}.$$

The first matches buyer-income means; the second matches EV-buyer income shares. Drop one bin per country because shares sum to one.

Sec. 4.3, p.16.

The US entrant construction is an extrapolation

$$\eta_{j,US} = \eta_{j,China} + \psi_g, \quad \psi_g = \mathbb{E}[\eta_{k,US} - \eta_{k,China} \mid k \text{ observed in both, } g].$$

Overlap: non-Chinese automakers' models sold in both the US and China; g : powertrain. Transfer $\xi_{j,China}$ and apply estimated US preferences.

Export-cost input	USD / vehicle
Shipping	2,750
US distribution and marketing	1,402

China–Europe cost analogues supply the export-market adjustment.

App. C.2, pp.61–63.

What the EU out-of-sample exercise tests

Three related samples, with distinct selection rules

Object	Coverage reported in the paper
Policy background	China-manufactured BEVs; CVD described by origin and brand.
Table 3 experiment	2023 Chinese-owned model set in six EU countries; includes BYD, MG, Aiways and Geely-group brands.
Synthetic control	Incumbent BEV/PHEV models of BYD, MG, Polestar, NIO and Lynk & Co; at least six positive-sales months in 2023.

Fit: 2023m1–2024m6 log sales. Donor weights: Volkswagen 70.3%, Fiat 24.0%, BMW 5.7%.

Model –56.0% versus –57.4% in the definitive-duty window.

Aggregate quantity validation; sample coverage is distinct from legal policy coverage.

Sec. 3.1, p.12; Table 3, p.43; App. D.1, p.67; Fig. E.5, p.74.

$$\Delta EB = -\Delta Q_{ICE} \times (\$3,810 + \$687),$$

$$\Delta CO_2 = -\Delta Q_{ICE} \times 78.5 \text{ tonnes.}$$

Recycling: 95.44 Mt of avoided lifetime CO₂ for one year's new-car cohort.

Grid mix, EV production emissions and driving responses are not separately modeled.

App. C.3, p.63; Table 4, p.44.

Manufacturing jobs are calibrated from sales

$$Jobs = \sum_{g \in \{US, TP\}} \rho_g (\ell_{ICE} Q_{g,ICE} + \ell_{EV} Q_{g,EV}).$$

TP: foreign-owned US transplants; ρ_g : domestic-production share.

Maintained input	US-owned	Transplants
Domestic-production share ρ_g	0.75	0.53
Labor intensity	ICE	EV
Jobs / vehicle	0.10	0.07

$$\ell_{EV} = 0.70 \ell_{ICE}, \quad \Delta Jobs_{recycling} = -51.95 \text{ thousand.}$$

Figure E.7 varies job value w while holding these coefficients fixed.

App. C.4, pp.63–65; Table 4, p.44.

Policy sets and fiscal baselines

Reported policy language, ownership, and fiscal baseline differ by experiment

	US Tables 2/4	EU Table 3	Cross-country Table 6
Taxed set	Ten added Chinese EVs	Existing Chinese-owned EV set	Common ten-entrant scenario
Subsidy wording	Domestically manufactured EVs / domestic EVs	EU-member-manufactured EVs	Domestic EVs; home-HQ counts used
Profit scope	US-headquartered firms	EU-headquartered automakers	Home-HQ automakers
Fiscal baseline	No entry / no policy; recycling $G = 0$	10% MFN benchmark; Table 3 reports $\Delta G = 0$ under recycling	Changes versus status quo

Ownership and production are distinct. Exact product-level eligibility mapping is not fully reported.

Tables 2–4, pp.42–44; Table 6, p.46; App. C.4, pp.63–64.

[◀ Return 4](#)

[◀ Return 17](#)

[◀ Return 21](#)

[◀ Return 26](#)

[◀ Return 29](#)

[◀ Return 30](#)

A fiscal-cost extension changes the objective

$$W_{\chi} = W - (\chi - 1) \max\{-G, 0\}, \quad \chi \geq 1.$$

Table F.5 re-evaluates previously selected policies; it does not re-optimize each policy for every χ .

$$\Gamma_0 = \{\gamma \in \Gamma : G(\gamma) = 0\} \subseteq \Gamma, \quad \max_{\gamma \in \Gamma_0} W_{\chi}(\gamma) \leq \max_{\gamma \in \Gamma} W_{\chi}(\gamma).$$

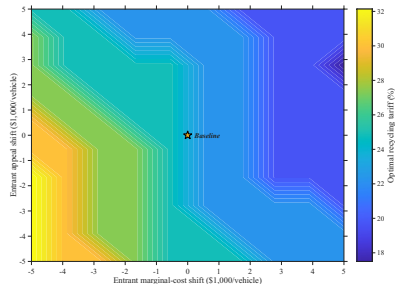
$\gamma = (\tau, b)$; Γ : joint-policy feasible set. Same objective and re-optimization on both sides.

App. C.5, pp.65–66; Table F.5, p.84.

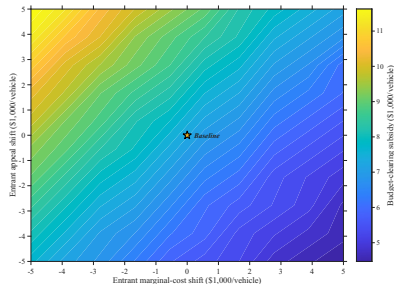
Sensitivity of the optimal policy instruments

Horizontal: entrant cost; vertical: entrant appeal; both shifts in \$1,000/vehicle

(c) Optimal recycling tariff



(d) Budget-clearing subsidy



Both instruments are re-optimized at each point. Higher appeal enlarges the fiscal base; the optimal tariff also depends on the welfare tradeoff. The star is the baseline.

App. D.2, p.68; Fig. E.10(c,d), p.79.

Joint policy paths and group prices

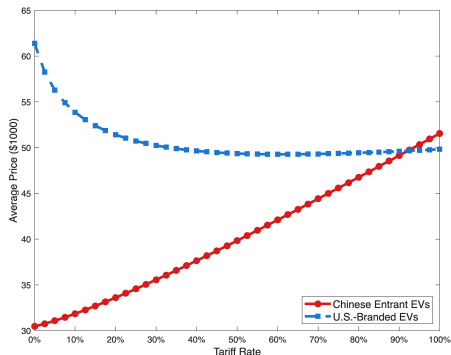


Figure 5(b) uses the paper's reported price series.

Fig. 5(b), p.39.

Joint policy path

$$\left. \frac{dp}{d\tau} \right|_{b(\tau)} = p_{\tau} + p_b b'(\tau).$$

Changing group weights

$$d\bar{p} = \sum_j w_j dp_j + \sum_j p_j dw_j.$$

w_j : model weights in the group average.

When is a zero policy instrument optimal?

Distinguish a policy boundary from a pricing equilibrium or an ineffective instrument

$$V = CS + \Pi_{\text{dom}} + EB, \quad \mathcal{L} = V + \lambda(G - \bar{G}) + \mu_{\tau}\tau + \mu_b b.$$

$$V_{\tau} + \lambda G_{\tau} + \mu_{\tau} = 0, \quad \mu_{\tau} \geq 0, \quad \mu_{\tau}\tau = 0,$$

$$V_b + \lambda G_b + \mu_b = 0, \quad \mu_b \geq 0, \quad \mu_b b = 0.$$

- **Interior tool:** its fiscal-weighted marginal welfare equals zero.
- **Zero tool:** that marginal value is nonpositive; do not impose the interior equality.
- **No eligible products:** the subsidy may be ineffective; choosing zero need not reflect a strictly negative marginal value.

Assumes $\tau, b \geq 0$ and constraint qualifications. These are necessary local conditions, not a global sufficiency theorem. Without a fiscal constraint, replace $V_{\gamma} + \lambda G_{\gamma}$ by W_{γ} .

Sec. 2.2, p.11; App. A, pp.47–52.

The full equilibrium welfare derivative

$D_p \mathbf{q} \in \mathbb{R}^{J \times J}$: demand Jacobian; $r \in \{\tau, b\}$: a scalar policy derivative

$$W(\tau, b) = CS + \Pi_{\text{dom}} + G + EB, \quad G = \tau \sum_{j \in \mathcal{C}} x_j q_j - b \sum_{j \in \mathcal{D}} q_j.$$

$$W_r = \underbrace{-\mathbf{q}^\top \mathbf{p}_r}_{\text{consumer surplus}} + \underbrace{\sum_{j \in J_{\text{dom}}} [q_j x_{j,r} + (x_j - c_j) q_{j,r}]}_{\text{domestic portfolio profits}} + \underbrace{G_r}_{\text{public funds}} + \underbrace{EB_r}_{\text{environment}}, \quad \mathbf{q}_r = (D_p \mathbf{q}) \mathbf{p}_r.$$

- $r \in \{\tau, b\}$: each policy changes every price and quantity.
- J_{dom} includes domestic firms' ICEs; foreign-owned profits are excluded.
- EB values lifetime damages avoided through the year's new-vehicle choices.

Sec. 2, pp.9–10; Sec. 5.1, pp.18–19.

Interior policy conditions: what must be quantified

$$R = \sum_{j \in \mathcal{C}} x_j q_j, \quad S = \sum_{j \in \mathcal{D}} q_j, \quad G = \tau R - bS, \quad V = W - G.$$

R : pretax import expenditure; S : eligible EV sales. For $r \in \{\tau, b\}$, the interior condition is $V_r + \lambda G_r = 0$.

$$\tau = \frac{V_\tau + \lambda R - \lambda b S_\tau}{-\lambda R_\tau}, \quad b = \frac{V_b + \lambda \tau R_b - \lambda S}{\lambda S_b}.$$

- Implicit conditions: prices, sales and derivatives depend on both policies.
- The tariff denominator is the import-expenditure slope, not revenue curvature.
- Signs require nonzero denominators and equilibrium response conditions; joint policy directions also depend on cross-curvature.

App. A.1, Props. 1–2, pp.47–49.

Estimation links shares to moment conditions

$$v_{ij} = \delta_j + \mu_{ij}; \quad \delta_j = X_j' \bar{\beta} + \eta_j + \xi_j.$$

1. Invert shares for each candidate preference vector θ :

$$s(\delta(\theta), \theta) = s^{obs} \implies \xi(\theta, \beta) = \delta(\theta) - X\beta - \eta.$$

2. Stack aggregate and buyer-income restrictions:

$$g(\theta, \beta) = \begin{pmatrix} N^{-1} \sum_{jmt} Z_{jmt} \xi_{jmt}(\theta, \beta) \\ m_{\text{model income}}(\theta) - m_{\text{model income}}^{obs} \\ m_{\text{EV income}}(\theta) - m_{\text{EV income}}^{obs} \end{pmatrix}, \quad (\hat{\theta}, \hat{\beta}) = \arg \min_{\theta, \beta} g' \widehat{\mathcal{W}} g.$$

3. Two-step GMM: obtain initial estimates, update moment weights, and re-estimate.

θ : all estimated tastes entering μ_{ij} , including $\bar{\alpha}$; $\beta = \bar{\beta}$: common attribute coefficients; η : nuisance fixed effects. N : observations; Z : instruments/exogenous attributes; g : stacked moments; $\widehat{\mathcal{W}}$: moment weights.

Sec. 4.3, p.16; Eqs. (4)–(5), pp.14–15.

Ten Chinese EV entrants

China, 2023 · Annual model sales in vehicles

Brand	Model	Sales
BYD	QINPLUS	327,371
BYD	DOLPHIN	285,970
BYD	YUANPLUS	282,501
BYD	SEAGULL	280,217
BYD	SONGPLUS	264,762
Wuling	WULINGBINGO	233,735
GAC Aion	AIONY	228,555
GAC Aion	AIONS	221,227
BYD	SONGPRO	188,007
MG	MG4/MGMULAN	152,337

Table F.2, p.81; Sec. 5.2, p.19.

Markups and prices

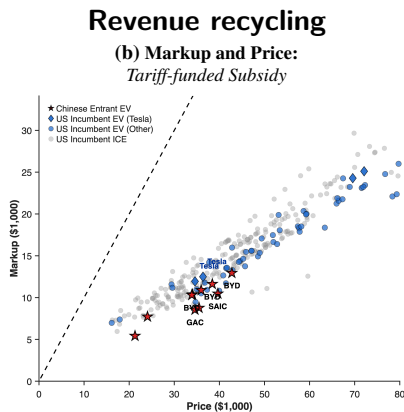
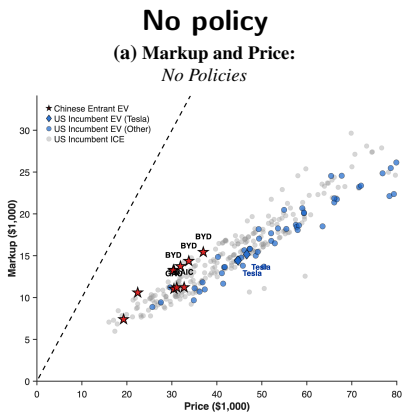


Fig. E.8(a,b), p.77.

Curvature and equilibrium positions

Authors' conceptual illustration

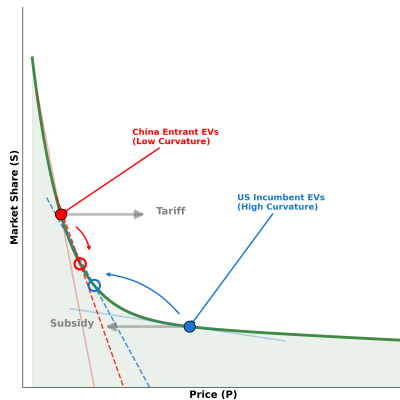


Fig. E.9, p.78.